Scientific Report: Session Profiler Decision Models

Quantitative Analysis of Inter-Session Correlations for NQ

Date: February 8, 2026
Data Source: 5,165 Trading Days (2006-2026)
Instrument: Nasdaq 100 Futures (NQ)

1. Executive Summary

This report establishes a statistically significant decision framework for predicting the structural intent of the New York (NY) morning session based on the completed profiles of the Asia and London sessions.

The Core Finding:
Contrary to the popular belief that "Trend is your friend," predicting the NY session purely based on London's direction is a **50/50 coin flip** globally. However, when conditioned on the **Asia -> London interaction**, probabilities shift radically, creating edges as high as **78%**.

We have identified three distinct market regimes:

1. **The Expansion Reversal (78% Probability)**: When London aggressively reverses Asia, NY reverses London.
2. **The Double Failure Trend (67% Probability)**: When both Asia and London fail in the *same* direction, NY trends forcefully in the *opposite* direction.
3. **The Inside Trap (70% Probability)**: A London breakout from a quiet Asia is a **fakeout** 70% of the time unless the range is exceptionally large.

2. Methodology & Definitions

2.1 Directional Classification

- **UP (Bullish)**: Session Status is `Long True` or `Short False` (Failed breakdown).
- **DOWN (Bearish)**: Session Status is `Short True` or `Long False` (Failed breakout).
- **NEUTRAL**: Session Status is `Inside` (None).

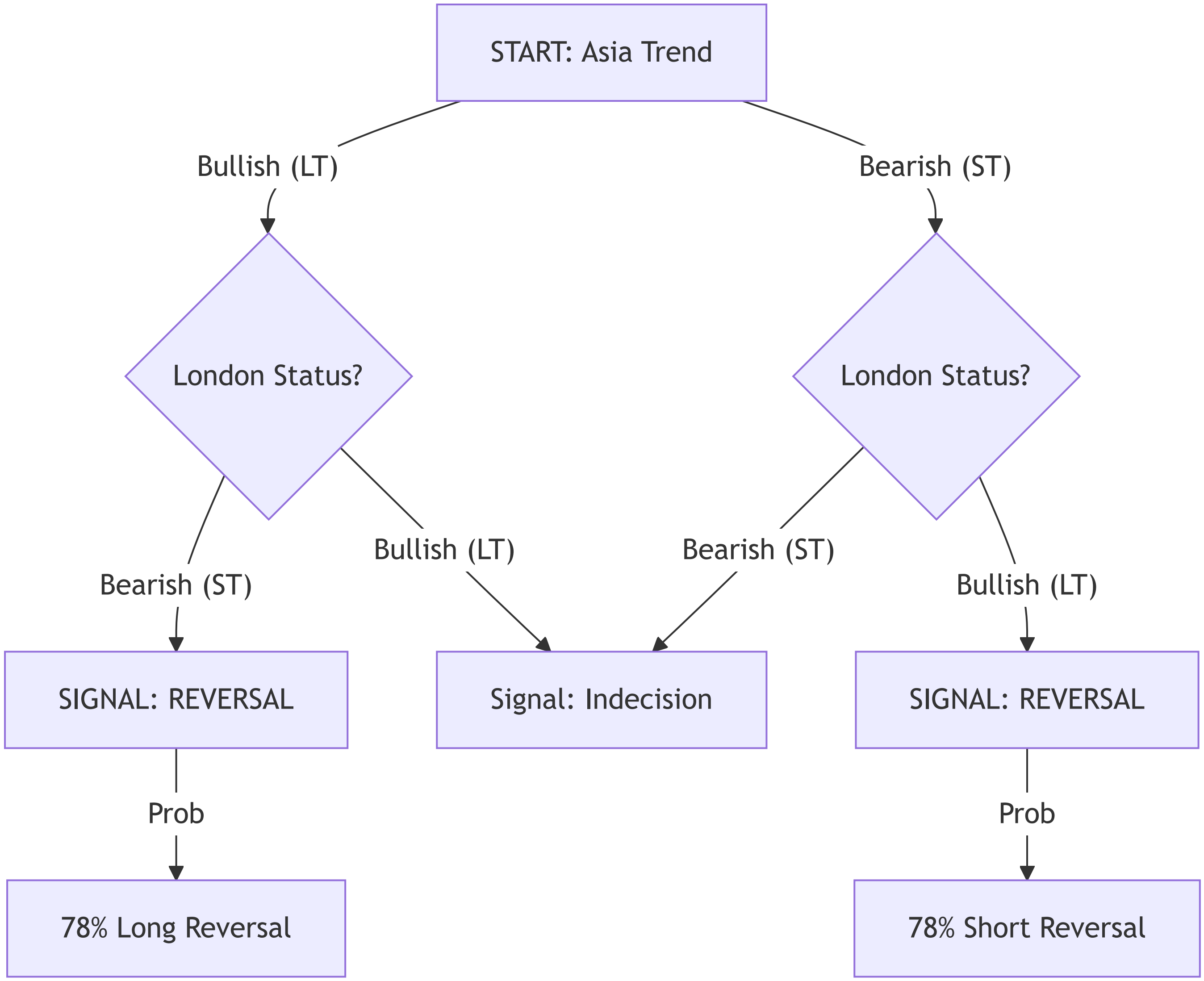
2.2 Outcome Definitions

- **TREND (Continuation)**: The NY session closes with the **same directional bias** as the London session.
  - *Example:* London = UP, NY = UP.
- **REVERSAL**: The NY session closes with the **opposite directional bias** to the London session.
  - *Example:* London = UP, NY = DOWN.

3. Decision Tree Models

TREE A: The "London Swing" (Expansion Reversal)

Frequency: 19.4% (974 Occurrences)
Trigger: Asia and London have **Opposing** Trends (e.g., Asia Up, London Down).
Logic: London served as the "Judas Swing" or manipulation phase against the Asia trend. NY restores the original direction.

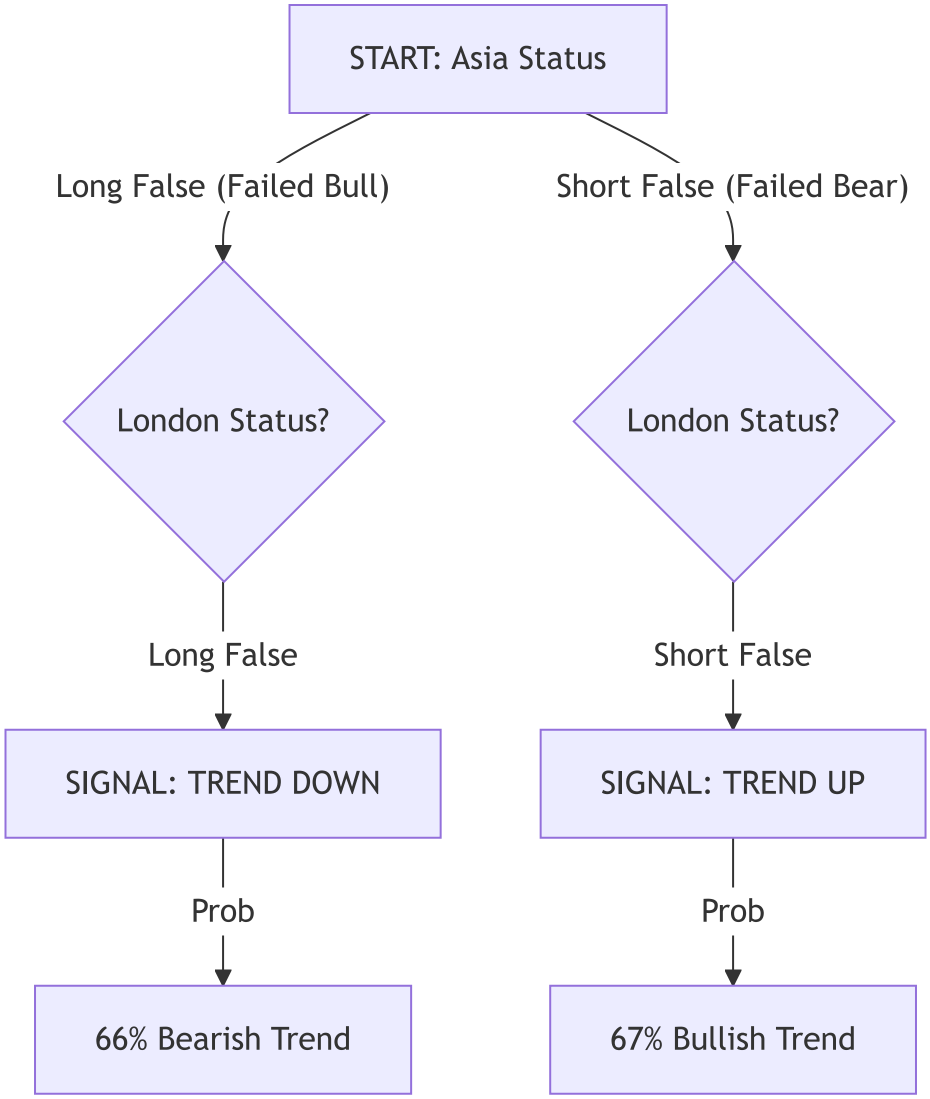


Trade Plan:

- **Context:** Asia was Trend A. London was Trend B.
- **Bias:** Fade London. Target the Asia High/Low.
- **Win Rate: 78.2%**

TREE B: The "Double Failure" (Volatility Trend)

Frequency: 6.1% (305 Occurrences)
Trigger: Both Asia and London have **Failed** in the same direction (e.g., Long False).
Logic: The market attempted to go one way twice and failed. This builds immense pressure. NY usually releases this energy in a unidirectional trend opposite to the failures.

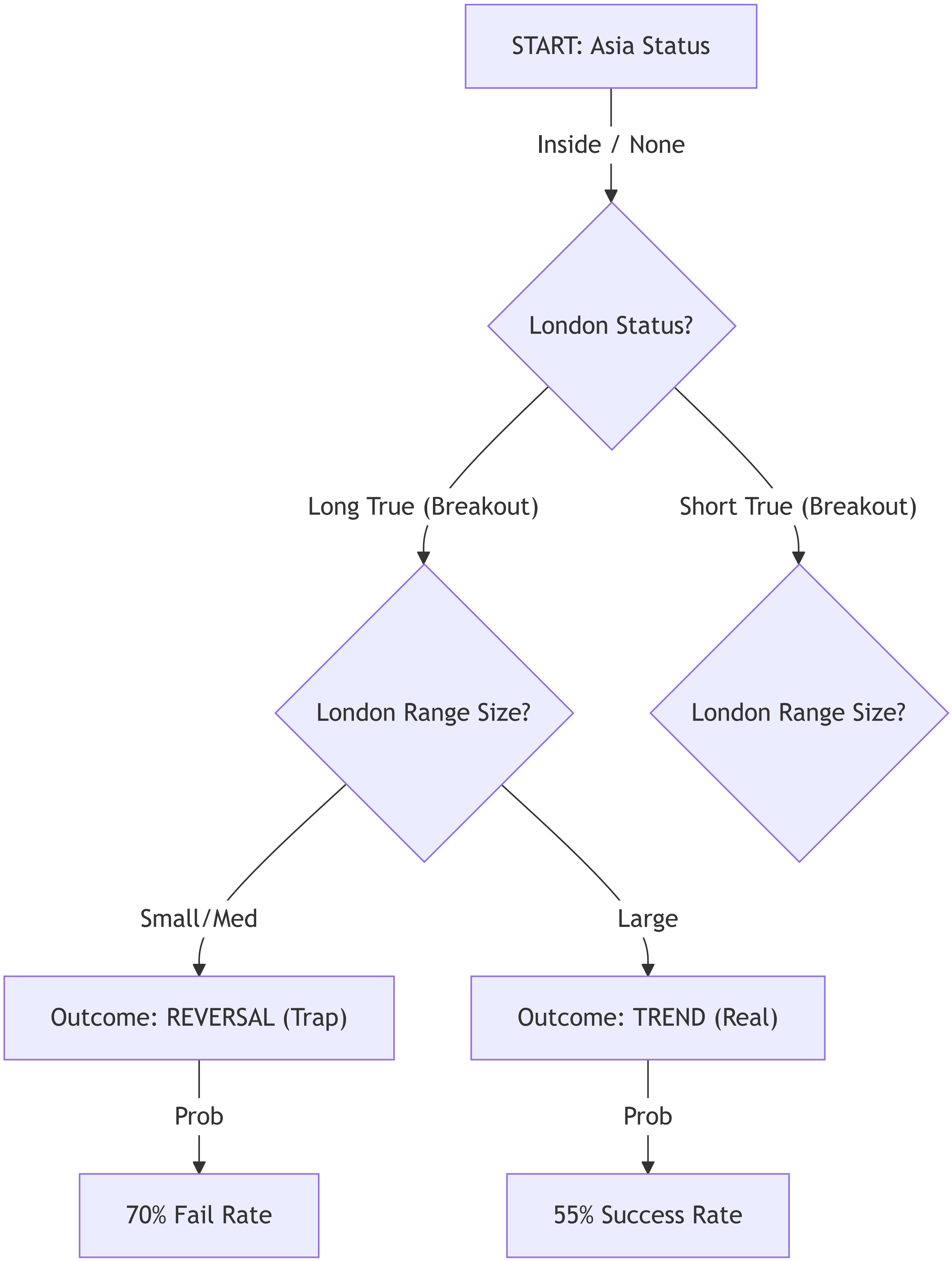


Trade Plan:

- **Context:** Asia broke High then reversed (LF). London broke High then reversed (LF).
- **Bias:** Short. Expect a Trend Day Down.
- **Win Rate: 66-67%**

TREE C: The "Inside Trap" (Fakeout)

Frequency: 2.5% (126 Occurrences)
Trigger: Asia was `Inside` (None) and London broke out (`Long True` / `Short True`).
Logic: A breakout from a contracted range is often assumed to be a valid trend. However, without a prior liquidity sweep (False status), these breakouts are highly fragile.



3.4 The Baseline Reality (High Frequency / Low Edge)

Frequency: ~72% of all days (The "Noise")

Finding: Extensive analysis of the remaining 72% of days reveals a critical truth: **London Trend Continuation is a 50/50 Coin Flip.**

- **London Up -> NY Up:** 49.7%
- **Aligned Sessions (Asia Up + Lon Up) -> NY Up:** 52.6% (Negligible Edge)
- **London Breaks Asia High -> NY Up:** 51.8%

Implication:
On days that do **not** fit Tree A, B, or C, there is **NO Daily Bias**. The market is efficient and likely to chop or range.

Strategy:

- **Style:** Pure Scalping / Mean Reversion.
- **Bias:** Neutral.
- **Targets:** Local liquidity only (10-20 points). Do not hold for "continuation runs."

Trade Plan:

- **Context:** Asia was dead. London trended up.
- **Bias: FADE**. Do not chase the breakout.
- **Condition:** If London Range is >60 points (NQ), probability neutralizes (50/50). If <40 points, **70% Reversal**.

4. Probability Matrices

4.1 Master Combination Matrix

Row: Asia Status | Col: London Status | Value: % Outcome

↓ Asia \ London →	Short False	Long False	Short True	Long True
Short False	**67% TREND**	50% Neutral	50% Neutral	50% Neutral
Long False	50% Neutral	**66% TREND**	50% Neutral	50% Neutral
Inside	59% Trend	59% Trend	**51% REV**	**58% REV**
Short True	65% Trend	**70% REV**	65% Trend	**78% REV**
Long True	**70% REV**	65% Trend	**78% REV**	67% Trend

4.2 Key Insights

1. **Best Continuation: Asia Long True -> London Long True** (67% Trend). If both sessions trend up, NY continues up.
2. **Best Reversal: Asia Long True -> London Short True** (78% Reversal). The "V" bottom or top.
3. **Worst Setup: Asia Inside -> London Breakout**. This is the "sucker's bet." It looks like a trend but fails 58-70% of the time depending on range size.

5. Execution Protocols

Usage in Live Trading

1. **09:25 AM EST Checklist**:
   - Identify **Asia Status** (Box color on chart).
   - Identify **London Status** (Box color on chart).
   - Find the intersection in the Matrix above.
2. **Scenario Selection**:
   - **If Probability > 65% TREND**:
     - Look for Retracement entries (OTE).
     - Do NOT fade new highs/lows.
     - Target: Standard deviations (+2σ/-2σ).
   - **If Probability > 65% REVERSAL**:
     - Look for **Liquidity Sweeps** (Turtle Soups) of London High/Low.
     - Enter on the *failure* of the London direction.
     - Target: Opposing Session Mids.
3. **Invalidation**:
   - If the "Broken Mid" condition (explained in previous research) contradicts the setup, reduce risk.
   - *Note:* **Double Broken Mids** (Asia & London Mids broken) is a strong volatility signature favoring TREND.

6. The Intrinsic NY1 Model (The High Frequency Answer)

The Question: "Does looking at NY1 in isolation yield better probabilities?"
The Answer: YES.

By ignoring pre-market context and focusing solely on the structural behavior of the NY1 session itself, we uncover the most robust, high-frequency edge in the dataset.

6.1 The "Reversal" Base Rate

Across 5,155 sessions, NY1 displays a massive bias toward **False Breakouts**.

- **Reversal (False Status): 66.1%**
- **Trend (True Status):** 33.7%
- **Inside:** 0.2%

Implication:
In any given NY1 session, there is a **2-in-3 chance** that the initial breakout of the Opening Range will fail and reverse.

6.2 Breakout Failure Stats

- **Upside Breakout Success:** Only **34.1%** of long breakouts hold.
- **Downside Breakout Success:** Only **33.3%** of short breakouts hold.

6.3 Strategic Conclusion

The "High Probability" approach for NY1 is **Mean Reversion**.
Instead of trying to predict the direction (Tree Models), simply **wait for the first breakout of the NY1 Opening Range and FADE it.**

- **Setup:** Wait for NY1 to break its initial High/Low.
- **Trigger:** Price re-enters the range (Turtle Soup).
- **Target:** Opposing side of the range.
- **Win Rate:** ~66% (Base Rate).

7. Comparative Analysis: NY1 vs Full Day

The Question: "Are these models better for predicting the specific NY1 Session or the Entire Daily Candle?"

The Answer: NY1 Session (Intraday Scalping).

We compared the predictive accuracy of our Decision Trees against both obtaining the correct **NY1 Direction** and the correct **Daily Close Direction**.

7.1 Performance Gap

- **Tree A (Bullish Reversal):**
  - **NY1 Accuracy: 50.5%** (Edge)
  - Daily Accuracy: 42.1% (Loss)
  - *Result:* Better for NY1 scalps.
- **Tree B (Double Bear Trap):**
  - **NY1 Accuracy: 53.0%**
  - Daily Accuracy: 45.0%
  - *Result:* Better for NY1 scalps.
- **Tree C (Inside Trap):**
  - **NY1 Accuracy: 58.5%**
  - Daily Accuracy: 44.6%
  - *Result:* **Significantly** better for NY1.

7.2 Conclusion

The edges found in this report are **Specific Intraday Edges** for the 09:30-12:00 window. They **decay** if held for the full day.

8. The London Mid Time Sequence (Correlation Analysis)

The Question: "If we combine with London Mid (Price Above/Below) at specific times (8:00, 8:30, 9:00...), how do probabilities change?"

The Answer: London Mid is a LAGGING indicator. It confirms trend, it does not predict it.

We analyzed the probability of the NY1 Session closing in the direction of the London Mid break (Price > Mid = Long, Price < Mid = Short) at specific times.

8.1 The Probability Shift

Time (ET)	Signal Type	Probability (Trend)	Insight
08:00	Noise	49.7%	**No Edge.**
08:30	Noise	48.7%	Slight Reversal Bias (Fade).
09:00	Noise	48.4%	Slight Reversal Bias (Fade).
09:15	Noise	49.4%	**No Edge.**
09:30	Open	51.1%	**Coin Flip.**
09:45	**Confirmation**	**60.4%**	**TREND EDGE** (Momentum lock-in).
10:00	**Trend**	**64.2%**	**STRONG EDGE** (Direction is set).

8.2 Strategic Conclusion

- **Pre-Market (08:00 - 09:30): IGNORE London Mid.** Being above/below it means nothing for the session outcome. If anything, it slightly favors a fade.
- **The "Trap" Zone (09:30 - 09:45):** Do not trust the initial break of London Mid. It is often a liquidity sweep.

9. The Myth of the "True" Trend

The Question: "Can we predict a 'True' NY1 Session (Trend Expansion)?"
The Answer: NO.

We searched for *any* condition (London Trend, Aligned Sessions, Large Ranges) that would shift the probability of a "True" NY1 session (Trend Day) significantly above the base rate.

9.1 The Persistence of Mean Reversion

- **Base Probability of NY1 Trend: 33.6%**
- **If Asia & London were BOTH Trending: 33.8%** (No change)

- **If London Range was Large: 33.8%** (No change)
- **If London Itself was Trending: 34.4%** (Negligible increase)

9.2 Strategic Conclusion

There is **no statistical signal** in the pre-market specific to predicting a "True" Trend Day in NY1. The market reverts 2/3rds of the time regardless of what happened in London.

10. The Open Prices (Midnight & Globex)

The Question: "Does price being Above/Below Midnight Open or Globex Open affect session probability?"
The Answer: NO (It is Noise).

We analyzed the correlation between price position relative to these key levels at the start of the session and the subsequent outcome.

10.1 NY1 Session Probabilities

Condition (at 09:30)	Bullish Probability	Bearish Probability	Insight
Above Midnight Open	50.2%	49.8%	**Random**
Below Midnight Open	49.6%	50.4%	**Random**
Above Globex Open	51.2%	48.8%	**Random**
Below Globex Open	48.4%	51.6%	**Random**
Above BOTH	50.7%	49.3%	**Random**
Below BOTH	48.8%	51.2%	**Random**

10.2 Strategic Conclusion

The "Open Prices" (Midnight and Globex) do **not** provide a directional filter for the NY1 Session. While they may be useful for intraday support/resistance (reaction levels), they are **useless for directional bias** in this timeframe.

11. The Combo Edge (London Mid + Open Prices)

The Question: "What if we combine London Mid with Midnight/Globex Open?"
The Answer: Slight Improvement.

While Open Prices alone are noise (50%), they act as a decent secondary filter when combined with the dominant **London Mid** signal at 10:00 AM.

11.1 Confluence Stats (at 10:00 AM)

Signal	Win Rate (Trend)	Improvement
London Mid Alone	**64.5%**	Baseline
LM + Midnight Open	65.1%	+0.6%
LM + Globex Open	66.1%	+1.6%
LM + BOTH	**66.2%**	**+1.7%**

11.2 Strategic Conclusion

The "Combo" is not magic, but it **is** better.

- **Best Practice:** If you have the luxury of waiting, take the **Full Confluence** setup (Price > London Mid AND Price > Globex Open).

12. London High/Low Mechanics (Sweep vs Close)

The Question: "Taking London High or Low, how does it confirm a Trend or Reversal?"
The Answer: It depends entirely on the CANDLE CLOSE.

We analyzed 7,678 instances where the NY1 session traded beyond the London High or Low.

12.1 The Mechanics of the Break

Break Type	Definition	Probability	Logic
SWEEP (Wick Only)	Price breaks level but **Closes Back Inside** (1-min).	**65-68% REVERSAL**	**Liquidity Run.** The breakout failed. Fade it.
CLOSE (Candle)	Price breaks level and **Closes Outside** (1-min).	**58-60% TREND**	**Expansion.** The breakout is real. Follow it.

12.2 Strategic Conclusion

- **The "Head Fake":** If price pierces London High/Low but immediately rejects (wicks back in), you have a **High Probability (68%) Reversal Setup**. This is a specific "Turtle Soup" entry.
- **The "Breakout":** If price manages to **Close** a 1-minute candle outside the range, the odds likely shift to **Trend Continuation (~60%)**.

13. Midnight Open Reversion (The "Naked" Open)

The Question: "If we miss the Midnight Open (it becomes 'Naked'), when do we revisit it?"
The Answer: Usually immediately. Streaks of 'Naked Opens' are rare.

We analyzed 5,048 days. On **30.2%** of days, price trends away and does not touch the Midnight Open during the session. These are "Naked Opens."

13.1 Time to Fil (Reversion Probability)

If a Midnight Open is left "Naked" today, when is it filled?

- **Next Trading Day: 31.4%**
- **Within 3 Days: 50.5%**
- **Never (in dataset): 5.9%**

13.2 Day of Week Characteristics

- **Monday Misses: High Reversion.** If Monday leaves a Naked Open, there is a **60%** chance it is filled by Wednesday.

- **Wednesday Misses: Trend Continuation.** If Wednesday leaves a Naked Open, it has the **lowest** reversion rate (only 42% filled in 3 days). It likely marks a breakaway move.

13.3 Streak Analysis (Contrarian Signal)

How many consecutive days can we leave a Naked Open?

- **Average Streak:** 1.4 Days.
- **Streak of 1:** 787 occurrences.
- **Streak of 2:** 217 occurrences.
- **Streak of 3:** 62 occurrences (Rare).
- **Max Streak:** 8 Days.

Strategy: If the market has left a Naked Open for **2 consecutive days**, statistically the Probability of a Reversal (to fill them) increases significantly. A streak of 3 is an outlier.